

Inverse Boundary Value Problems for the Jordan–Moore–Gibson–Thompson Equation

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- **Introduction**

Physical background, Kuznetsov-type model, Westervelt-type model

- **Main uniqueness results**

Local Well-posedness, Recovery of β in the Westervelt-type model, Recovery of (β, κ) in the Kuznetsov-type model

- **Proof strategy**

Second order linearization, Integral identities, Geometric optics solutions, Gaussian beam solutions, Injectivity of geodesic ray transform

- **Summary**

Conclusions, Future work

Introduction

Physical background of the JMGT equation

- The Jordan–Moore–Gibson–Thompson (JMGT) equation is a third-order nonlinear acoustic model for high-intensity sound propagation in thermoviscous media.
- It can be viewed as a relaxation-type modification of the classical Westervelt equations, $\partial_t^2 u - c^2 \Delta u - \partial_t \Delta u = \partial_t^2 (\beta u^2)$.
- The resulting third-order-in-time hyperbolic model incorporates quadratic acoustic nonlinearity, thermoviscous damping, and relaxation effects, making it relevant for high-intensity ultrasound applications.

A typical JMGT equation with Kuznetsov-type nonlinearity is

$$\tau \partial_t^3 u + \alpha \partial_t^2 u - c^2 \Delta u - b \Delta \partial_t u = \partial_t \left(\beta (\partial_t u)^2 + \kappa |\nabla u|^2 \right).$$

Here

- $\tau > 0$ is the thermal relaxation time, α is the frictional damping coefficient, $b > 0$ is the sound diffusivity, and $c > 0$ is the sound speed;
- β describes the quadratic acoustic nonlinearity, while κ describes the nonlinear effect induced by spatial gradients of the acoustic field.

The JMGT equation with Kuznetsov-type nonlinearity

We study the JMGT equation on a compact Riemannian manifold (M, g) with boundary:

$$\mathcal{M} = (0, T) \times M, \quad \Gamma = (0, T) \times \partial M.$$

- The Riemannian metric g provides an effective geometric description of an inhomogeneous and anisotropic acoustic medium.

We consider the Kuznetsov-type JMGT equation

$$\begin{cases} \partial_t^3 u + \alpha \partial_t^2 u - b \Delta_g \partial_t u - c^2 \Delta_g u = \partial_t (\beta(x)(\partial_t u)^2 + \kappa(x)|\nabla_g u|_g^2), & \text{in } \mathcal{M}, \\ u = f, & \text{on } \Gamma, \\ u(0, \cdot) = \partial_t u(0, \cdot) = \partial_t^2 u(0, \cdot) = 0, & \text{in } M. \end{cases}$$

The target nonlinear coefficients are

$$\beta \quad \text{and} \quad \kappa.$$

The JMGT equation with Westervelt-type nonlinearity

We also consider a simplified JMGT model with Westervelt-type nonlinearity. In classical nonlinear acoustics, the Westervelt equation is often formulated as a pressure-based model, while the Kuznetsov equation is commonly written in terms of the acoustic velocity potential.

The Westervelt-type JMGT equation is

$$\begin{cases} \partial_t^3 u + \alpha \partial_t^2 u - b \Delta_g \partial_t u - c^2 \Delta_g u = \partial_t^2 (\beta(x) u^2), & \text{in } \mathcal{M}, \\ u = f, & \text{on } \Gamma, \\ u(0, \cdot) = \partial_t u(0, \cdot) = \partial_t^2 u(0, \cdot) = 0, & \text{in } M. \end{cases}$$

The target nonlinear coefficient is

$$\beta.$$

Dirichlet-to-Neumann maps and inverse problems

The Dirichlet-to-Neumann (DtN) map sends the prescribed boundary data f to the measured Neumann-type boundary flux of the corresponding solution.

For the Kuznetsov-type model, we define the corresponding DtN map $\Lambda_{\beta,\kappa}^K$ as

$$\Lambda_{\beta,\kappa}^K : f \mapsto (c^2 \partial_\nu u + b \partial_\nu \partial_t u) \Big|_\Gamma.$$

Here ν is the outward unit normal vector to ∂M with respect to g , and $\partial_\nu u := \langle \nabla_g u, \nu \rangle_g$.

For the Westervelt-type model, we define the corresponding DtN map Λ_β^W as

$$\Lambda_\beta^W : f \mapsto (c^2 \partial_\nu u + b \partial_\nu \partial_t u) \Big|_\Gamma.$$

Inverse boundary value problems

The inverse problems we consider are the following:

- Can we recover β from the DtN map Λ_β^W ?
- Can we recover β and κ from the DtN map $\Lambda_{\beta,\kappa}^K$?

Main uniqueness results

Function spaces

For a nonnegative integer m , define

$$H^m(\Gamma) = \bigcap_{k=0}^m H^k([0, T]; H^{m-k}(\partial M)),$$

$$A_m = \bigcap_{k=0}^{m+1} C^k([0, T]; H^{m+\frac{3}{2}-k}(\partial M)) \cap H^{m+2}(\Gamma),$$

$$N_{m+1} = \bigcap_{k=1}^{m+1} H^k([0, T]; H^{m+1-k}(\partial M)).$$

We work in the energy space

$$E^m = \bigcap_{k=0}^m C^k([0, T]; H^{m-k}(M)),$$

with

$$\|u\|_{E^m} := \sup_{t \in [0, T]} \sum_{k=0}^m \|\partial_t^k u(\cdot, t)\|_{H^{m-k}(M)}.$$

Local well-posedness for small boundary data and initial data

Proposition

Assume that $m > n + 1$. Fix

$$\beta, \kappa \in C^m(M), \quad \beta = \kappa = 0 \quad \text{on } \partial M.$$

There exists a constant $\delta > 0$ such that if

$$(v_0, v_1, v_2, f) \in \mathcal{N}_\delta \subset H^{m+2}(M) \times H^{m+1}(M) \times H^m(M) \times A_m,$$

where

$$\mathcal{N}_\delta = \left\{ (v_0, v_1, v_2, f) : \|v_0\|_{H^{m+2}(M)} + \|v_1\|_{H^{m+1}(M)} + \|v_2\|_{H^m(M)} + \|f\|_{H^{m+2}(\Gamma)} < \delta, \right. \\ \left. \text{and } (v_0, v_1, v_2, f) \text{ satisfy compatibility conditions up to order } m+1 \right\},$$

then the Kuznetsov-type JMGT equation admits a unique solution

$$u \in E^{m+2}, \quad \partial_\nu u \in N_{m+1},$$

and

$$\|u\|_{E^{m+2}} + \|\partial_\nu u\|_{N_{m+1}} \leq C \left(\|v_0\|_{H^{m+2}(M)} + \|v_1\|_{H^{m+1}(M)} + \|v_2\|_{H^m(M)} + \|f\|_{H^{m+2}(\Gamma)} \right).$$

Main theorem: Euclidean setting

Let $M := \Omega \subset \mathbb{R}^n$, $n \geq 2$, be a bounded connected domain with smooth boundary, and set

$$\Gamma = [0, T] \times \partial\Omega, \quad \gamma = \alpha - \frac{c^2}{b} > 0.$$

Theorem (Uniqueness in the Euclidean setting)

Assume that α, b, c are constants,

$$T > \frac{1}{\sqrt{b}} \operatorname{diam}(\Omega), \quad (v_0, v_1, v_2) = (0, 0, 0),$$

$$(\beta_1, \kappa_1), (\beta_2, \kappa_2) \in C^m(\Omega) \times C^m(\Omega), \quad \beta_j = \kappa_j = 0 \quad \text{on } \partial\Omega, \quad j = 1, 2.$$

If there exists $\delta > 0$ such that

$$\Lambda_{\beta_1, \kappa_1}^K(f) = \Lambda_{\beta_2, \kappa_2}^K(f)$$

for all $f \in A_m$ with $(0, 0, 0, f) \in \mathcal{N}_\delta$, then

$$\beta_1 = \beta_2, \quad \kappa_1 = \kappa_2 \quad \text{in } \Omega.$$

Main theorem: Riemannian Westervelt-type case

Let $\dim M = 3$ and let $\alpha, b, c \in C^\infty(M)$, we impose the following geometric assumptions.

Geometric setting

We consider the conformal metric $b^{-1}g$, which determines the effective propagation geometry. We assume that

$$T > \text{diam}_{b^{-1}g}(M),$$

and that ∂M is strictly convex with respect to $b^{-1}g$, while $(M, b^{-1}g)$ satisfies the foliation condition, that is, there exists a smooth strictly convex function $f : M \rightarrow \mathbb{R}$.

Theorem (Westervelt-type JMGT equation)

Under the Geometric setting, let $\Lambda_{\beta_1}^W$ and $\Lambda_{\beta_2}^W$ denote the DtN maps for the Westervelt-type JMGT equation with coefficients

$$\beta_1, \beta_2 \in C^m(M), \quad \beta_j = 0 \quad \text{on } \partial M, \quad j = 1, 2.$$

If there exists $\delta > 0$ such that

$$\Lambda_{\beta_1}^W(f) = \Lambda_{\beta_2}^W(f)$$

for all $f \in A_m$ with $(0, 0, 0, f) \in \mathcal{N}_\delta$, then

$$\beta_1 = \beta_2 \quad \text{in } M.$$

Main theorem: Riemannian Kuznetsov-type case

Theorem (Kuznetsov-type JMGT equation)

Under the above geometric setting, let $\Lambda_{\beta_1, \kappa_1}^K$ and $\Lambda_{\beta_2, \kappa_2}^K$ denote the DtN maps for the Kuznetsov-type JMGT equation with coefficients

$$(\beta_1, \kappa_1), (\beta_2, \kappa_2) \in C^m(M) \times C^m(M),$$

and assume that

$$\beta_j = \kappa_j = 0 \quad \text{on } \partial M, \quad j = 1, 2.$$

If there exists $\delta > 0$ such that

$$\Lambda_{\beta_1, \kappa_1}^K(f) = \Lambda_{\beta_2, \kappa_2}^K(f)$$

for all $f \in A_m$ with $(0, 0, 0, f) \in \mathcal{N}_\delta$, then

$$\beta_1 = \beta_2, \quad \kappa_1 = \kappa_2 \quad \text{in } M.$$

Earlier work: nonlinear hyperbolic equations

- [Kurylev–Lassas–Uhlmann, 2018]: For scalar wave equations with quadratic nonlinearity on globally hyperbolic Lorentzian manifolds, local measurements determine the conformal class of the Lorentzian metric. This work introduced the use of nonlinear wave interactions in inverse problems.
- [Feizmohammadi–Oksanen, 2022]: For nonlinear wave equations, $\square_g u + Vu = u^3$, the source-to-solution map determine zeroth-order coefficient V .
- [Hintz–Uhlmann–Zhai, 2022]: For semilinear wave equations on Lorentzian manifolds, $\square_g u + au^4 = 0$, the Dirichlet-to-Neumann map determines nonlinear coefficients and the metric up to a natural obstruction.
- [Oksanen–Salo–Stefanov–Uhlmann, 2024]: A unified framework was developed for inverse problems associated with general real principal type differential operators.

Earlier work: Physically relevant models

- [Acosta–Uhlmann–Zhai, 2022]: For a Westervelt equation arising in nonlinear ultrasound imaging, $\partial_t^2 u - c^2 \Delta u = \partial_t^2 (\beta(x) u^2)$, the DtN map uniquely determines the nonlinear coefficient β .
- [Li–Zhang, 2024]: For a more general Westervelt equation, $\partial_t^2 u - \Delta u - \partial_t \Delta u + q(t, x)u = \partial_t^2 (\beta(t, x) u^2)$, the DtN map determines the potential q and the nonlinear coefficient β .
- [Jiang–Liu–Ni–Zhang, 2025]: For nonlinear progressive wave equations arising in infrasonic wave propagation, $\partial_t^2 u - \Delta(c_1 u + c_2 u^m) = 0$, the determination of c_1 and c_2 was studied.
- [Fu–Yao–Yu, 2024; Fu–Yu, 2025]: For the JMGT equation of Westervelt-model, inverse problems for recovering time-dependent nonlinear coefficient β from all boundary measurement was considered, and ibvp for the MGT equation was considered.
- [Kaltenbacher, 2025]: For the frequency-domain formulation of the JMGT model based on a multiharmonic expansion, local uniqueness and reconstruction of the spatial acoustic nonlinearity parameter and the sound speed were investigated.
- [Kaltenbacher–Rundell, 2023]: For the Westervelt equation, simultaneous reconstruction of the nonlinearity coefficient and the sound speed was considered.

Our contribution

Compared with previous works, our contribution is twofold.

- We prove a simultaneous recovery result for the two quadratic nonlinear coefficients

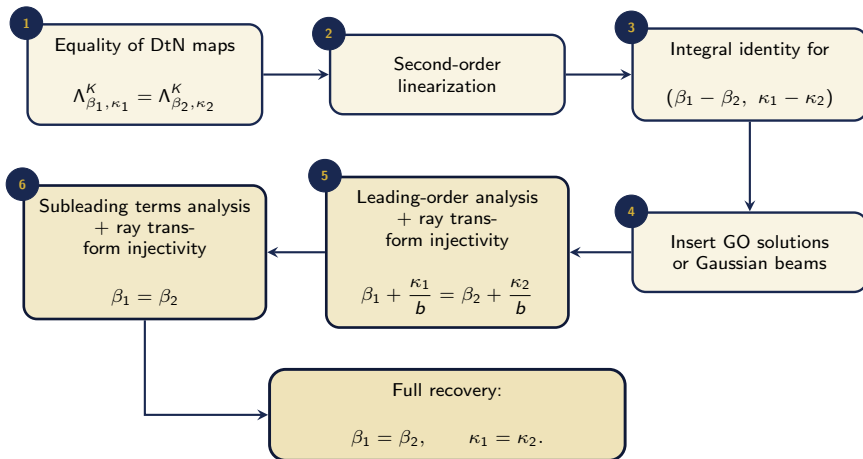
$$\beta \quad \text{and} \quad \kappa$$

in the Kuznetsov-type JMGT equation.

- We extend the uniqueness results to compact Riemannian manifolds under suitable geometric assumptions.

Proof of main theorem

Proof structure



How to recover nonlinear coefficients from the DtN map (1)

The recovery is based on the second order linearization method.

For small parameters $\varepsilon_1, \varepsilon_2$, take

$$f = \varepsilon_1 f_1 + \varepsilon_2 f_2$$

and let $u_{\varepsilon_1, \varepsilon_2}$ be the corresponding solution.

Using the first order linearization, we set

$$u_j = \left. \frac{\partial u_{\varepsilon_1, \varepsilon_2}}{\partial \varepsilon_j} \right|_{\varepsilon_1 = \varepsilon_2 = 0}, \quad j = 1, 2.$$

Then u_j solves the linear MGT equation

$$\begin{cases} \partial_t^3 u_j + \alpha \partial_t^2 u_j - b \Delta_g \partial_t u_j - c^2 \Delta_g u_j = 0, & \text{in } Q, \\ u_j = f_j, & \text{on } \Gamma, \\ u_j|_{t=0} = \partial_t u_j|_{t=0} = \partial_t^2 u_j|_{t=0} = 0, & \text{in } M. \end{cases}$$

How to recover nonlinear coefficients from the DtN map (2)

Define the second order linearization

$$w = \left. \frac{\partial^2 u_{\varepsilon_1, \varepsilon_2}}{\partial \varepsilon_1 \partial \varepsilon_2} \right|_{\varepsilon_1 = \varepsilon_2 = 0}.$$

Since the boundary data is linear in $\varepsilon_1, \varepsilon_2$, we have

$$w = 0 \quad \text{on } \Gamma,$$

and w has zero initial data.

For the Kuznetsov-type model, w satisfies

$$\begin{cases} \partial_t^3 w + \alpha \partial_t^2 w - b \Delta_g \partial_t w - c^2 \Delta_g w \\ \quad = 2 \partial_t (\beta \partial_t u_1 \partial_t u_2 + \kappa \langle \nabla_g u_1, \nabla_g u_2 \rangle_g), & \text{in } Q, \\ w = 0, & \text{on } \Gamma, \\ w|_{t=0} = \partial_t w|_{t=0} = \partial_t^2 w|_{t=0} = 0, & \text{in } M. \end{cases}$$

How to recover nonlinear coefficients from the DtN map (3)

Let (β_1, κ_1) and (β_2, κ_2) have the same DtN map, and set

$$\delta\beta = \beta_1 - \beta_2, \quad \delta\kappa = \kappa_1 - \kappa_2.$$

Let u_0 solve the adjoint linear MGT equation

$$\begin{cases} \mathcal{L}^* u_0 = 0, & \text{in } Q, \\ u_0 = h, & \text{on } \Gamma, \\ u_0|_{t=T} = \partial_t u_0|_{t=T} = \partial_t^2 u_0|_{t=T} = 0, & \text{in } M, \end{cases}$$

where, since α, b, c depend on x ,

$$\mathcal{L}^* u_0 = -\partial_t^3 u_0 + \alpha(x) \partial_t^2 u_0 + \partial_t \Delta_g (b(x) u_0) - \Delta_g (c^2(x) u_0).$$

Using integration by parts, we obtain

$$\int_0^T \int_M (\delta\beta \partial_t u_1 \partial_t u_2 + \delta\kappa \langle \nabla_g u_1, \nabla_g u_2 \rangle_g) \partial_t u_0 \, dV_g \, dt = 0.$$

We only need to recover β and κ from the above identity.

Geometric optics solutions in the Euclidean setting

For the linearized MGT equation

$$Pu = \partial_t^3 u + \alpha \partial_t^2 u - b \Delta \partial_t u - c^2 \Delta u = 0,$$

we construct geometric optics solutions.

WKB ansatz

$$u_\lambda(t, x) = e^{i\lambda\varphi(t, x)} (a_0(t, x) + \lambda^{-1} a_1(t, x)) + R_\lambda(t, x), \quad \lambda \gg 1.$$

Take

$$\varphi(t, x) = t + \psi(x).$$

Substituting the ansatz into $Pu = 0$, the leading term is

$$i\lambda^3 (b|\nabla\psi|^2 - 1) (a_0 + \lambda^{-1} a_1).$$

Thus the phase is determined by

$$b|\nabla\psi|^2 = 1.$$

In the Euclidean setting, choose

$$\psi(x) = \frac{1}{\sqrt{b}} \omega \cdot x, \quad \omega \in \mathbb{S}^{n-1}.$$

Geometric optics solutions in the Euclidean setting

Transport equations

Recall that

$$\gamma = \alpha - \frac{c^2}{b}.$$

The principal and subprincipal amplitudes satisfy

$$\left(\partial_t - \sqrt{b} \omega \cdot \nabla + \frac{\gamma}{2} \right) a_0 = 0,$$

$$\left(\partial_t - \sqrt{b} \omega \cdot \nabla + \frac{\gamma}{2} \right) a_1 = \tilde{\xi}(a_0).$$

The principal amplitude can be chosen explicitly as

$$a_0(t, x) = e^{-\frac{\gamma}{2}t} \phi_\epsilon(x + \sqrt{b} t \omega - p), \quad p \notin \bar{\Omega}.$$

Here ϕ_ϵ is a localized cutoff, chosen so that the initial support lies outside Ω . Hence a_0 , and similarly a_1 , satisfies the zero initial conditions in Ω .

The remainder R_λ solves a forced linear MGT equation and satisfies

$$\|R_\lambda\|_{L^2(Q)} + \|\partial_t R_\lambda\|_{L^2(Q)} + \|\nabla R_\lambda\|_{L^2(Q)} \longrightarrow 0, \quad \lambda \rightarrow \infty.$$

From GO solutions to ray transform identities

Starting from the integral identity

$$\int_Q \partial_t u_0 (\delta\beta \partial_t u_1 \partial_t u_2 + \delta\kappa \nabla u_1 \cdot \nabla u_2) dx dt = 0,$$

we insert the GO solutions

$$u_j = e^{i\lambda\varphi}(a_{j,0} + \lambda^{-1}a_{j,1}) + R_{j,\lambda}, \quad u_0 = e^{-2i\lambda\varphi}(a_{0,0} + \lambda^{-1}a_{0,1}) + R_{0,\lambda}.$$

Dividing by λ^3 and letting $\lambda \rightarrow \infty$, we obtain

$$\int_Q \left(\delta\beta + \frac{\delta\kappa}{b} \right) a_{1,0} a_{2,0} a_{0,0} dx dt = 0.$$

Take localized amplitudes

$$a_{j,0} = e^{-\frac{\gamma}{2}t} \phi_\epsilon(x + \sqrt{b}t\omega - p), \quad a_{0,0} = e^{\frac{\gamma}{2}t} \phi_\epsilon(x + \sqrt{b}t\omega - p).$$

Then, letting $\epsilon \rightarrow 0$, we get

$$\int_0^T \left(\delta\beta + \frac{\delta\kappa}{b} \right) (p - \sqrt{b}t\omega) e^{-\frac{\gamma}{2}t} dt = 0.$$

By injectivity of the attenuated X-ray transform in the Euclidean space, we obtain

$$\delta\beta + \frac{\delta\kappa}{b} = 0 \quad \text{in } \Omega.$$

From GO solutions to ray transform identities

Next, we consider the terms of order λ^2 . After dividing the integral identity by λ^2 and letting $\lambda \rightarrow \infty$, the terms involving

$$\delta\beta + \frac{\delta\kappa}{b}$$

vanish. Hence the remaining identity reduces to

$$-2\gamma \int_Q e^{-\frac{\gamma}{2}t} \delta\beta \phi_\epsilon^3(x + \sqrt{b}t\omega - p) dx dt = 0.$$

Letting $\epsilon \rightarrow 0$, we obtain the attenuated X-ray transform

$$\int_0^T \delta\beta(p - \sqrt{b}t\omega) e^{-\frac{\gamma}{2}t} dt = 0.$$

By injectivity of the attenuated X-ray transform,

$$\delta\beta = 0 \quad \text{in } \Omega.$$

Together with

$$\delta\beta + \frac{\delta\kappa}{b} = 0,$$

we conclude that

$$\delta\beta = 0, \quad \delta\kappa = 0.$$

Gaussian beam solutions for the JMGT equation

Consider the linear MGT equation

$$Qu := \partial_t^3 u + \alpha \partial_t^2 u - b \Delta_g \partial_t u - c^2 \Delta_g u = 0 \quad \text{in } \mathcal{M} = (0, T) \times M.$$

Let $\bar{g} = -dt^2 + b^{-1}g$. We construct Gaussian beams concentrated near a null geodesic $\varsigma(\tau) = (\tau, \sigma(\tau))$, where σ is a unit-speed geodesic in $(M, b^{-1}g)$.

Gaussian beam ansatz

In Fermi coordinates (τ, z') near ς , with $z' = 0$ on the ray,

$$u_\rho(t, x) = e^{i\rho\varphi(t, x)} a_\rho(t, x) + \mathcal{R}_\rho(t, x), \quad \rho \gg 1,$$

where

$$a_\rho = \chi\left(\frac{|z'|}{\delta}\right) \sum_{k=0}^{N+1} \rho^{-k} a_k(\tau, z').$$

The complex phase satisfies the eikonal equation up to high order: $\langle d\varphi, d\varphi \rangle_{\bar{g}} = \mathcal{O}(|z'|^{N+1})$, and has the local form

$$\varphi(\tau, z') = r + \sum_{i,j=1}^3 H_{ij}(\tau) z^i z^j + \dots, \quad \Im H(\tau) > 0.$$

The principal amplitude is determined by the transport equation. Along the central ray,

$$a_{0,0}|_\varsigma = (\sqrt{b(\tau)} \det Y(\tau))^{-\frac{1}{2}} \exp\left(-\frac{\sqrt{2}}{4} \int_{\tau_-}^{\tau} \gamma(s, 0, 0, 0) ds\right), \quad \gamma = \alpha - \frac{c^2}{b}.$$

Gaussian beams and weighted ray transforms

Choose Gaussian beams concentrated near the same null geodesic ς :

$$u_1 = u_2 = e^{i\rho\varphi} a_\rho + \mathcal{R}_\rho, \quad u_0 = e^{-2i\rho\bar{\varphi}} \bar{a}_{2\rho}^{(0)} + R_{2\rho},$$

where

$$a_\rho = \chi_\delta \sum_{k=0}^{N+1} \rho^{-k} a_k, \quad \|\mathcal{R}_\rho\|_{C^2} + \|R_{2\rho}\|_{C^2} = O(\rho^{-3}).$$

Step 1: leading-order term

After substitution and stationary phase in the transverse variables, the highest-order term gives

$$\int_\sigma b^{-\frac{1}{4}} \left(\delta\beta + \frac{\delta\kappa}{b} \right) \boxed{e^{-\frac{\sqrt{2}}{4} \int_{\tau-}^{\tau} \gamma(s) ds} (\det Y(\tau))^{-\frac{1}{2}}} d\tau = 0.$$

By injectivity of the single weighted ray transform, $\delta\beta + \frac{\delta\kappa}{b} = 0$.

Step 2: next-order term

Using $\delta\beta + \delta\kappa/b = 0$, the $\rho^{1/2}$ -order term yields

$$\int_\sigma b^{-\frac{1}{4}} \partial_\tau(\delta\beta) \boxed{e^{-\frac{\sqrt{2}}{4} \int_{\tau-}^{\tau} \gamma(s) ds} (\det Y(\tau))^{-\frac{1}{2}}} d\tau = 0.$$

Again by injectivity and $\delta\beta|_{\partial M} = 0$, we obtain $\delta\beta = 0$, $\delta\kappa = 0$.

Summary

Conclusion

- 1 We study inverse boundary value problems for the JMGT equation with quadratic nonlinearities.
- 2 The DtN map uniquely determines β in the Westervelt-type model.
- 3 The DtN map uniquely determines (β, κ) in the Kuznetsov-type model.
- 4 The proof combines second order linearization with GO solutions, Gaussian beams, and ray transform injectivity.
- 5 The results hold both in Euclidean domains and on compact Riemannian manifolds under geometric assumptions.

- **The remaining case $\gamma \neq 0$.** In the Euclidean case, our method can also fit the condition $\gamma < 0$. However, when $\gamma = 0$, it fails.
- **Boundary determination in the JMGT equation**
- **Stability estimates.** Establish quantitative stability results for the recovery of the nonlinear coefficients in the JMGT equation.
- **Reconstruction algorithms.** Develop constructive recovery procedures based on the integral identities obtained from higher order linearization. This would provide a bridge from uniqueness theory to numerical reconstruction.

Thank you for your attention!

Questions?